

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

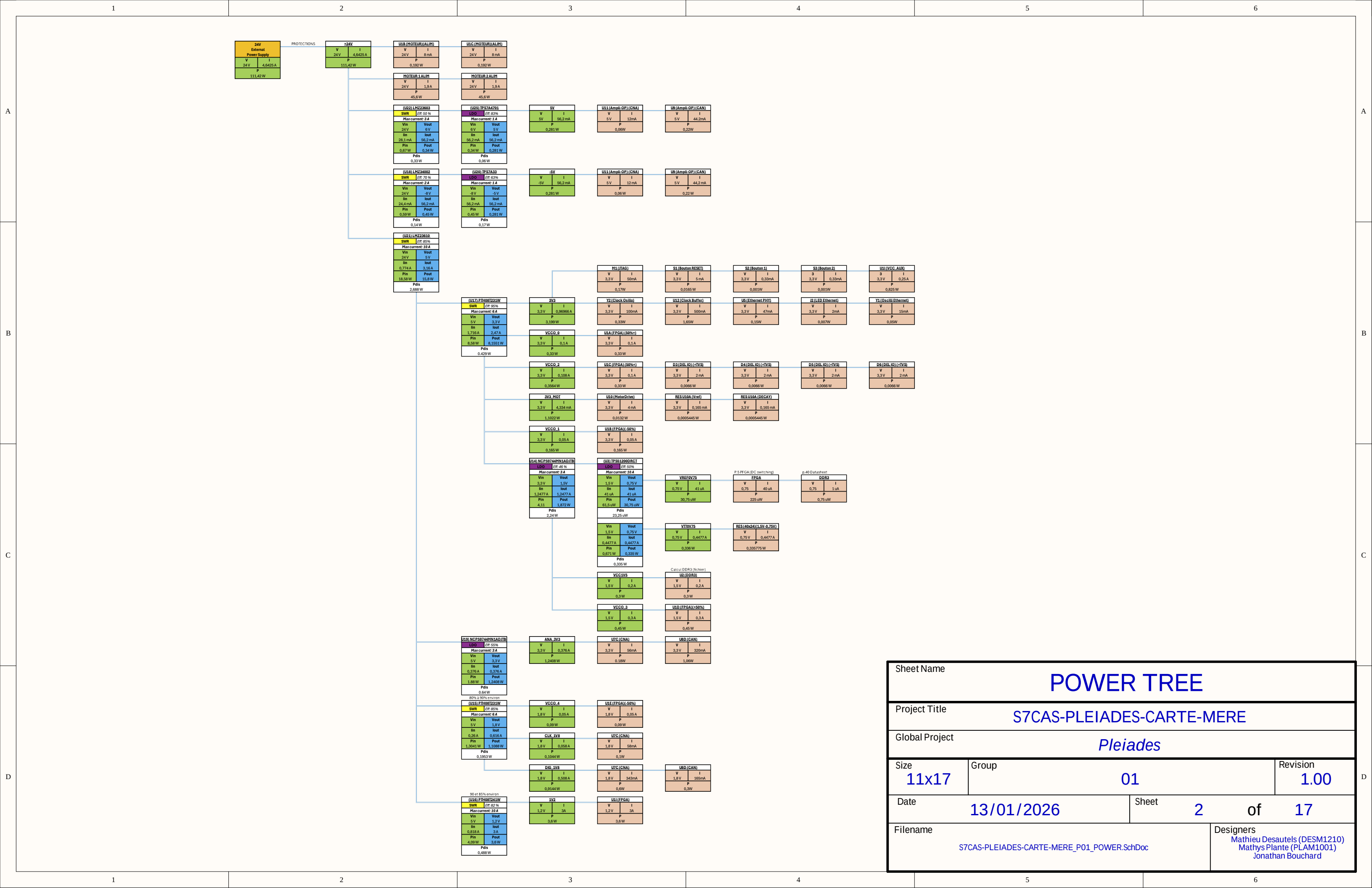
Date : 13/01/2026

Revision history	

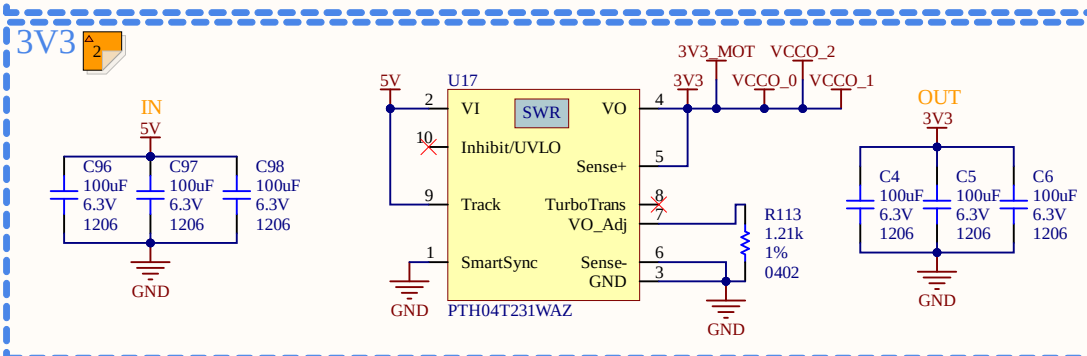
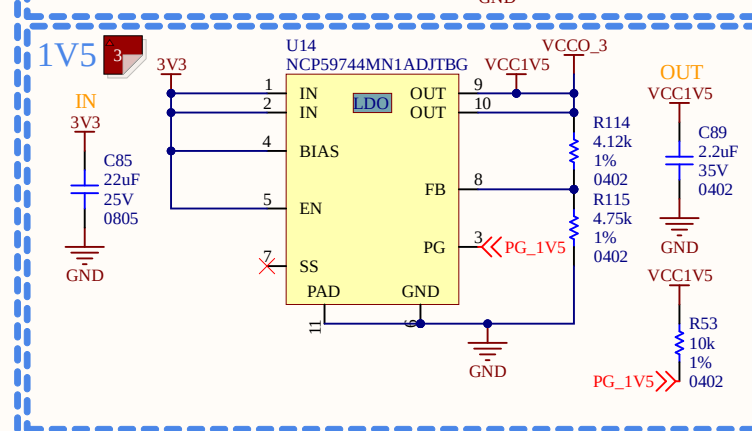
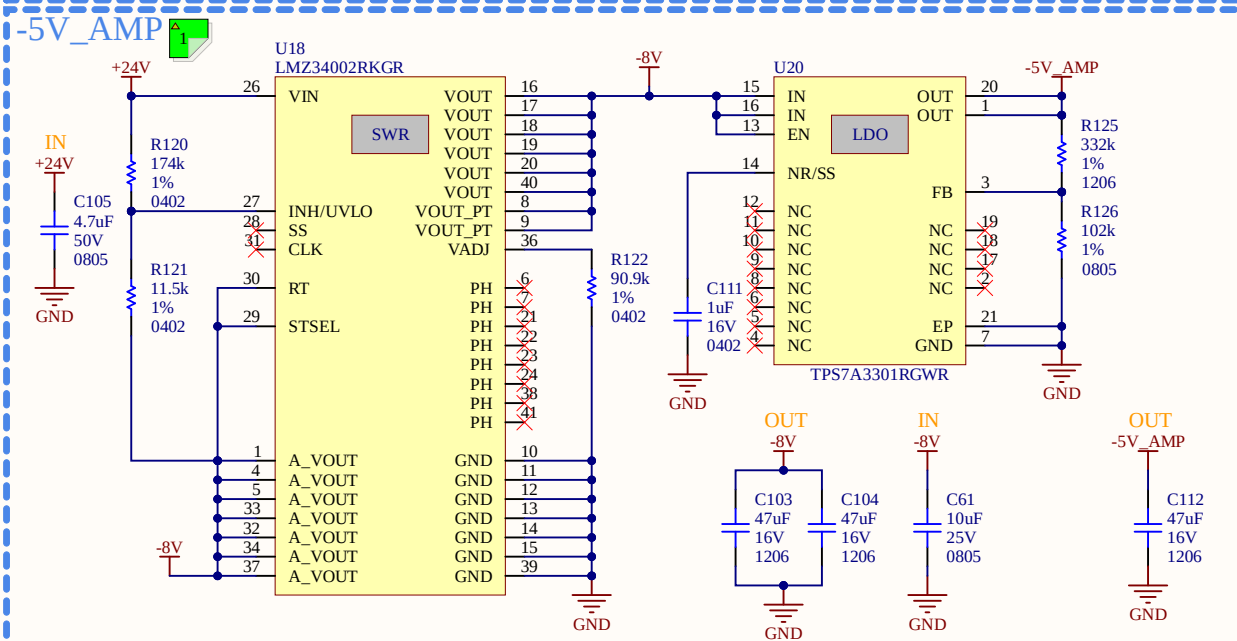
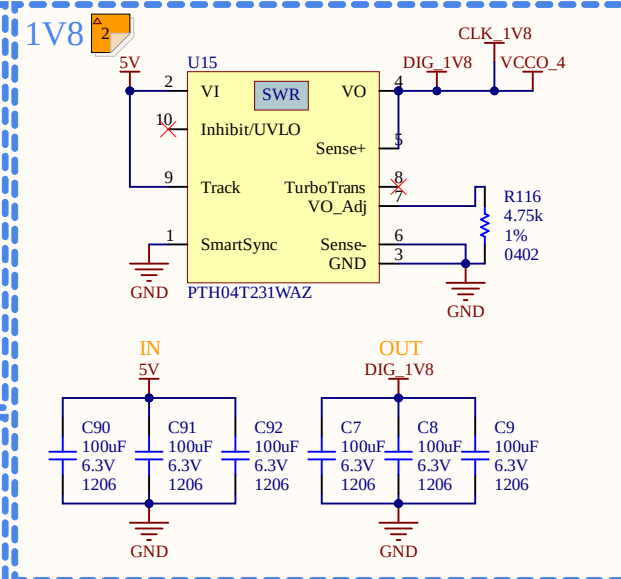
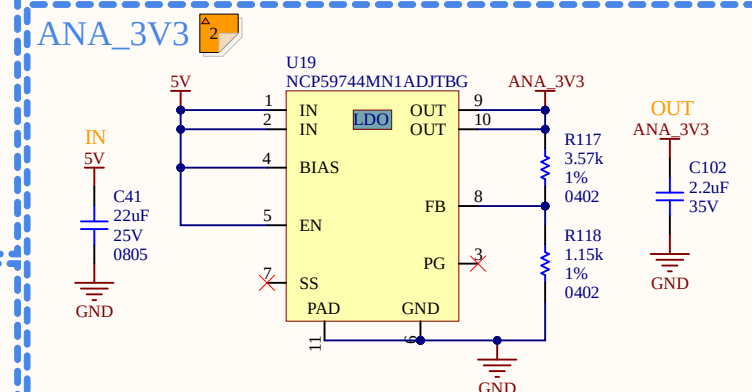
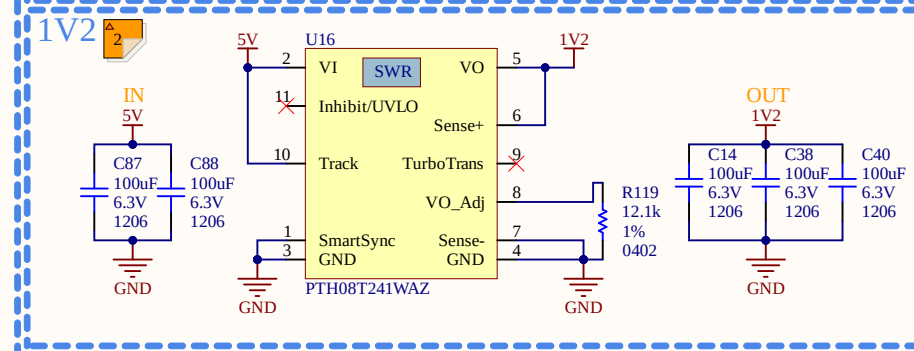
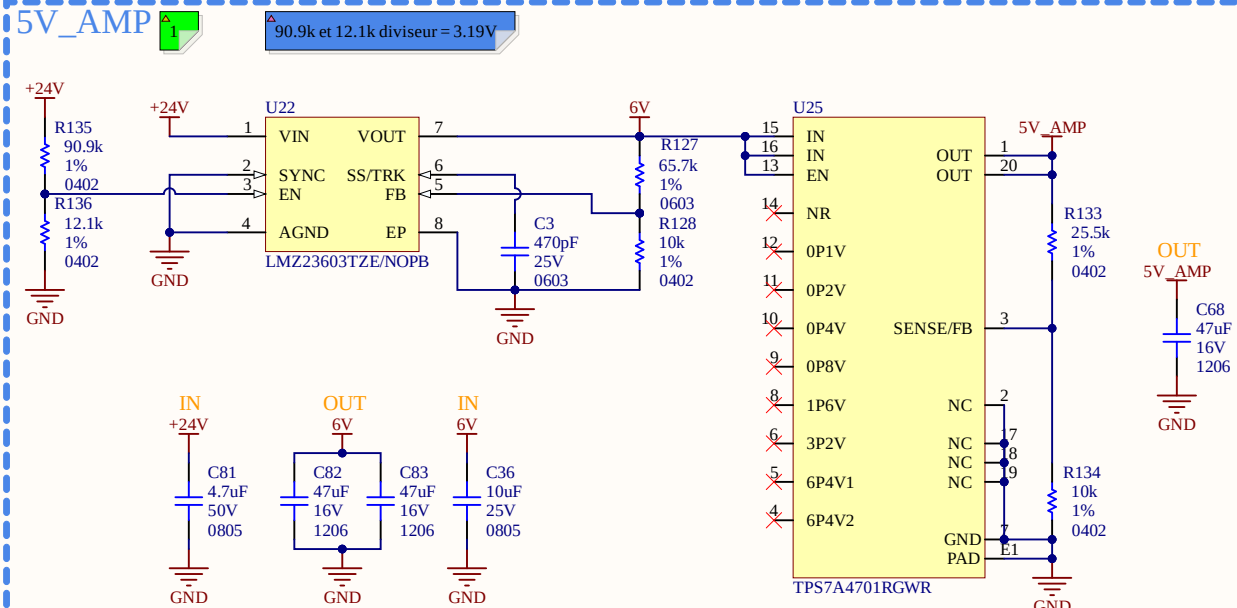
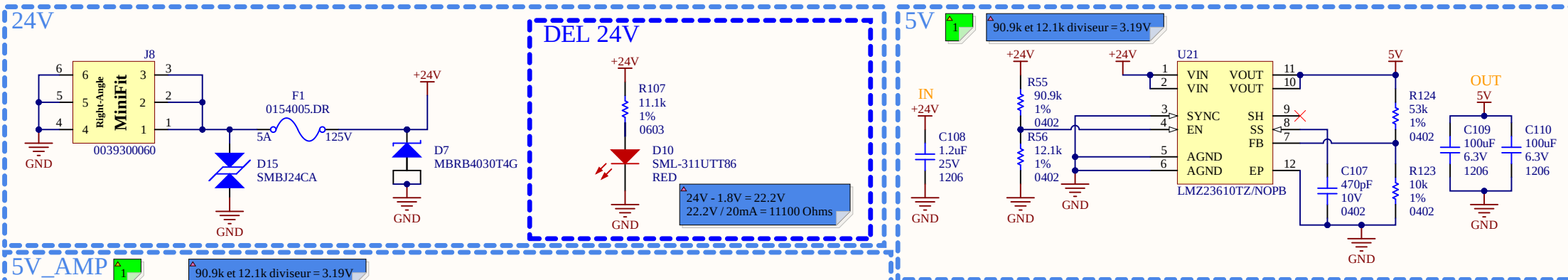
Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512



Project Title		
S7CAS-PLEIADES-CARTE-MERE		
Global Project		
Pleiades		
Size	Group	Revision
11x17	01	1.00
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13/01/2026	1 of 17	
Filename		Designers
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

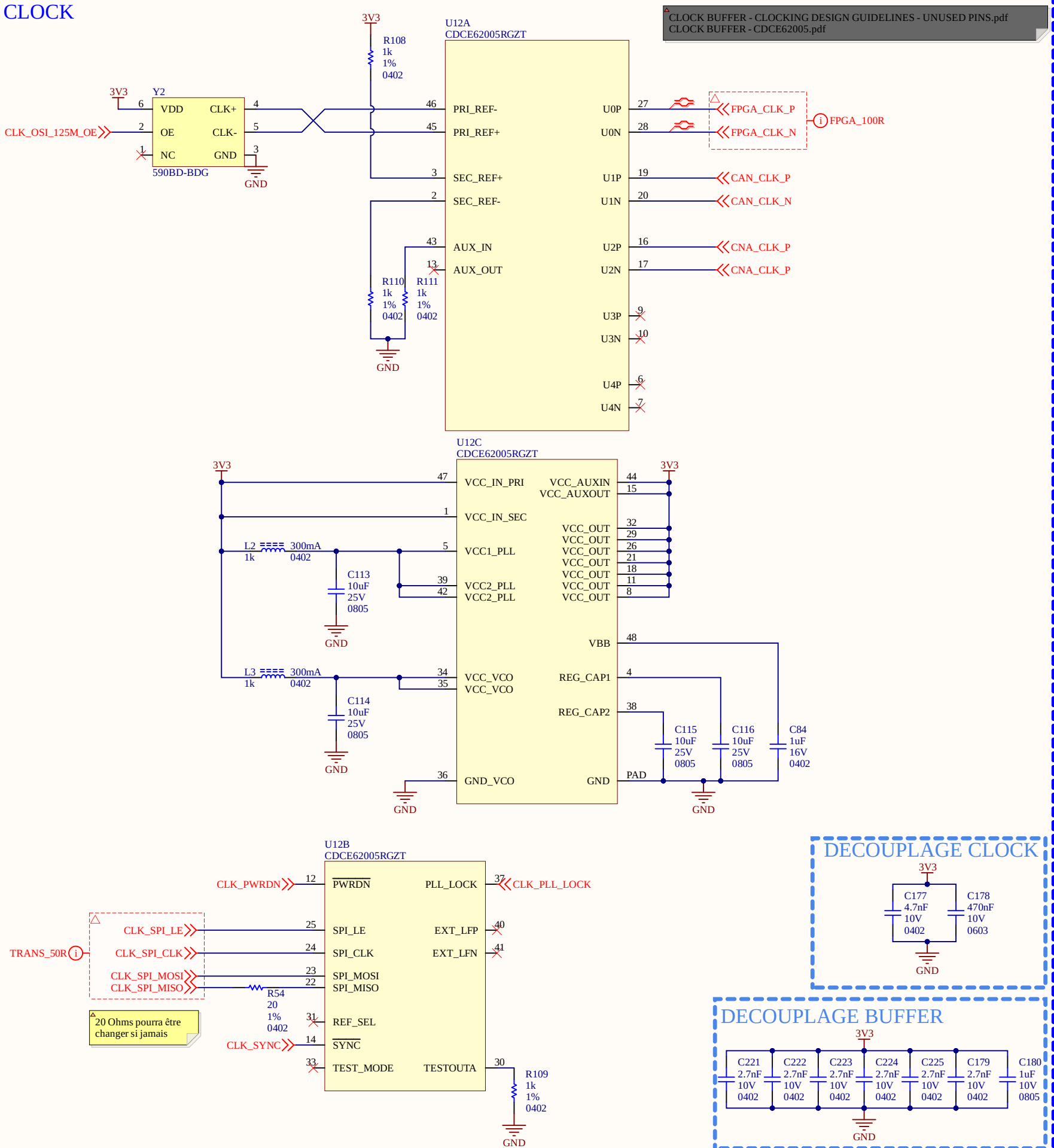


Sheet Name		POWER TREE	
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Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	2 of 17
Filename		Designers	
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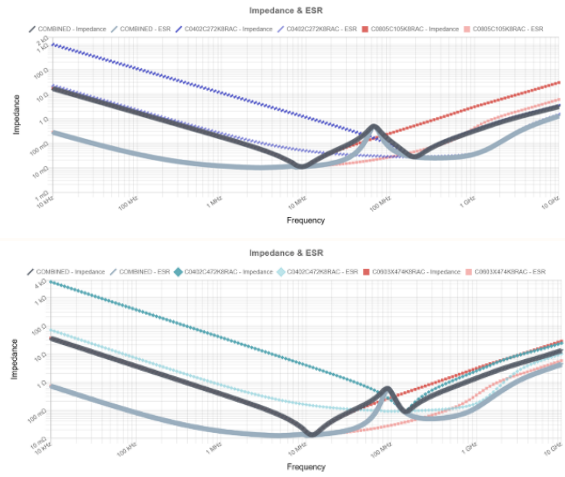


Sheet Name			ALIM		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	3	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_ALIM.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CLOCK



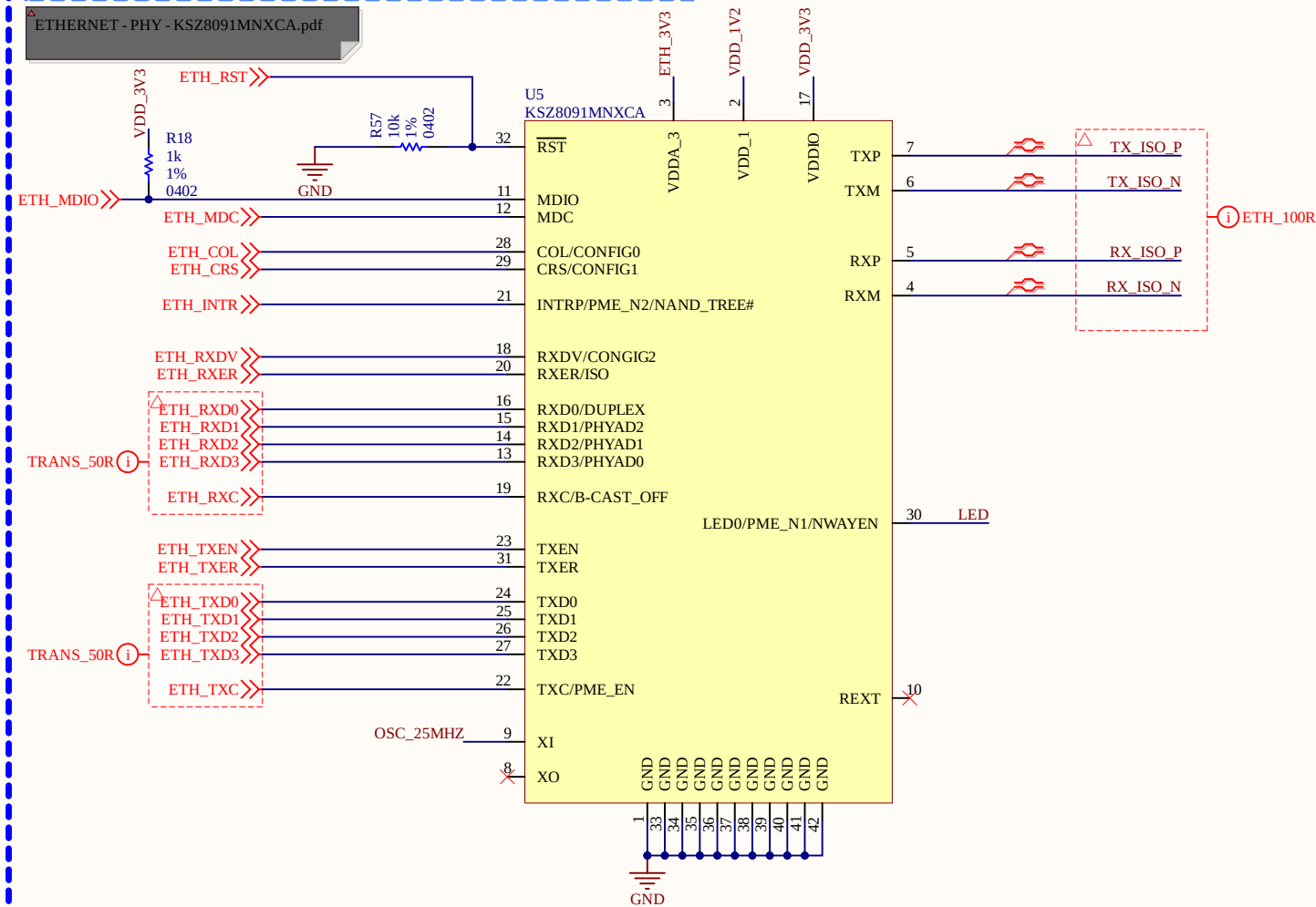
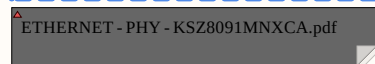
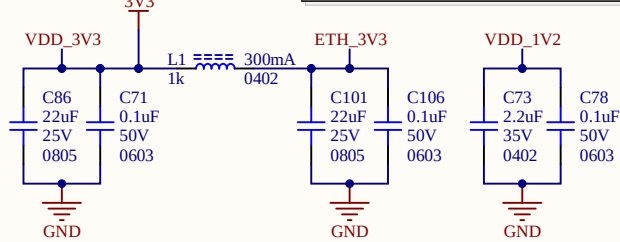
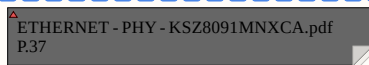
CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf
CLOCK BUFFER - CDCE62005.pdf



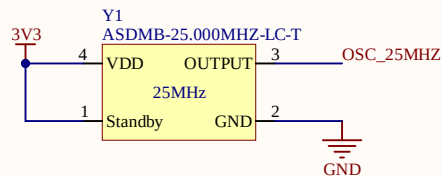
-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)
-CAN et CNA pas fait pour la pair différentiel puisqu'elle est directement dans le CAN et CNA

Sheet Name			CLOCK		
Project Title			S7CAS-PLIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17		01		1.00	
Date		13/01/2026		Sheet	4 of 17
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			Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

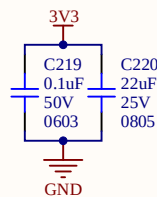
ETHERNET ALIMENTATION



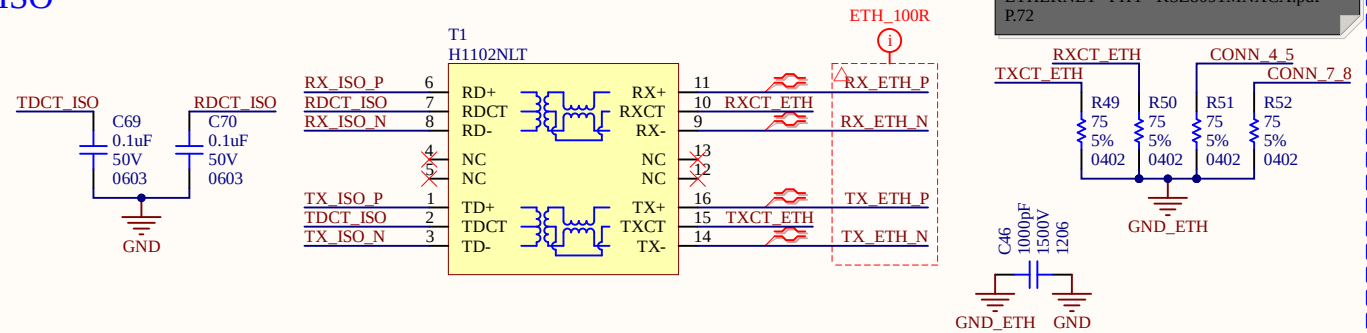
OSCILLI



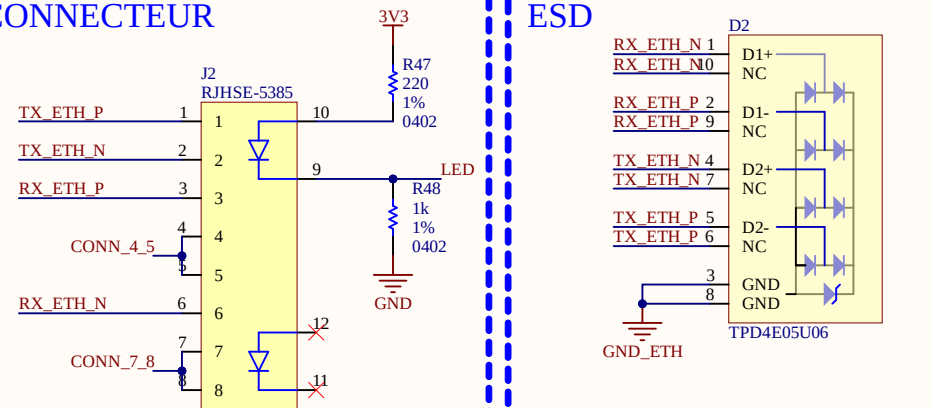
DECOUPLAGE



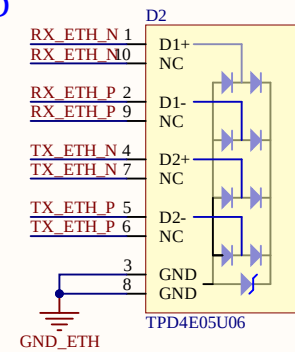
ISO



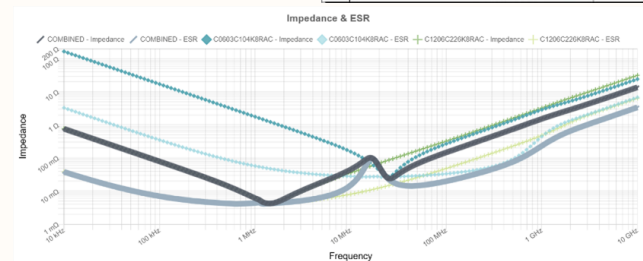
CONNECTEUR



ESD



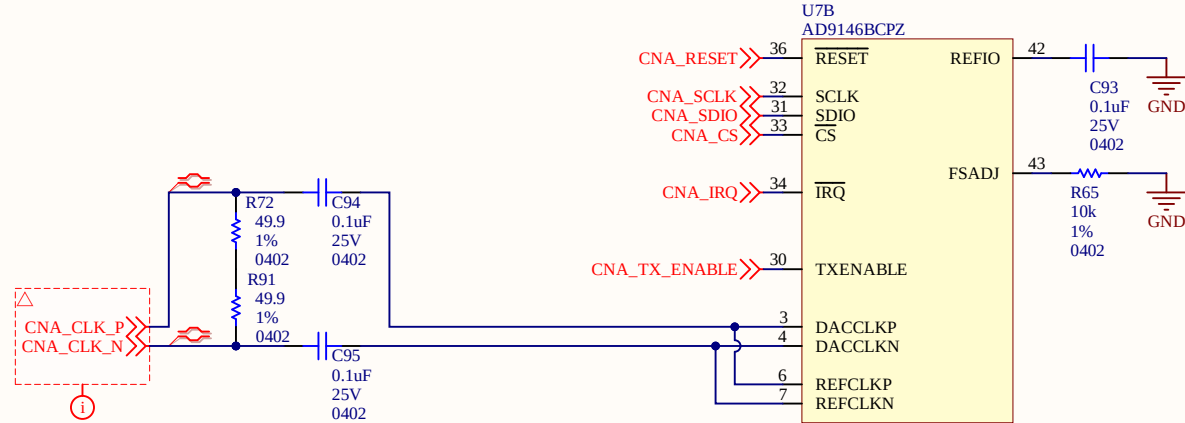
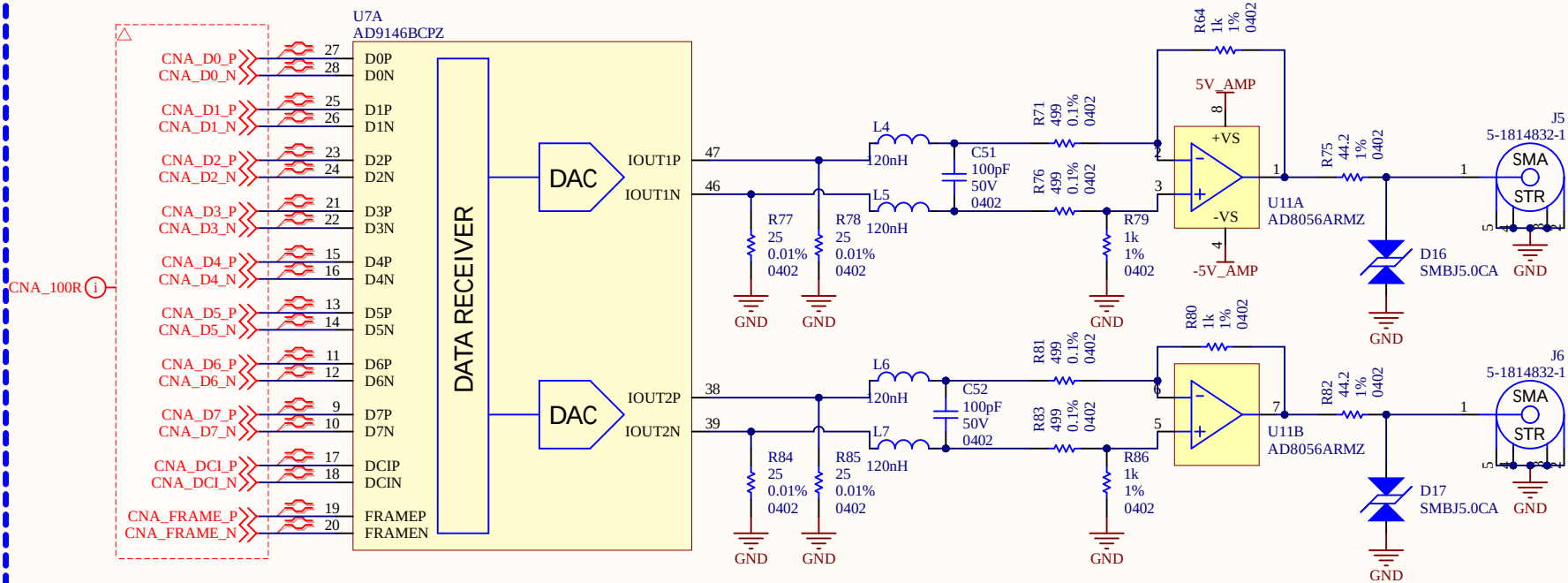
TIA 568A			Signal
Pin #	Wire Color Legend		
1		White/Green	TX+
2		Green	TX-
3		White/Orange	RX+
4		Blue	TRD2+
5		White/Blue	TRD2-
6		Orange	RX-
7		White/Brown	TRS3+
8		Brown	TRD3-



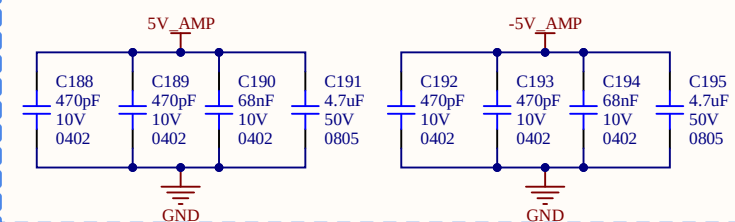
- Isolation du ETHERNET 1.5kV
- Ajout d'un condensateur de 1.5kV
- Ligne de transmission de 50R
- TX et RX voir le référence design pourquoi c'est ligne de transmission sans différentiel

Sheet Name			ETHERNET		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
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Filename			Designers		
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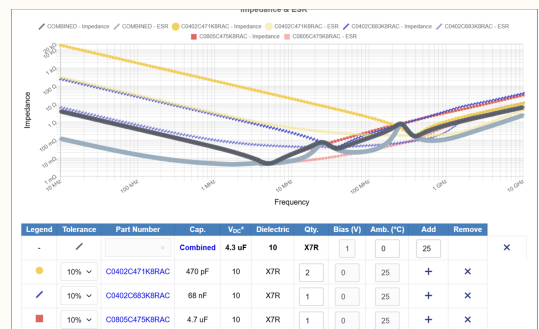
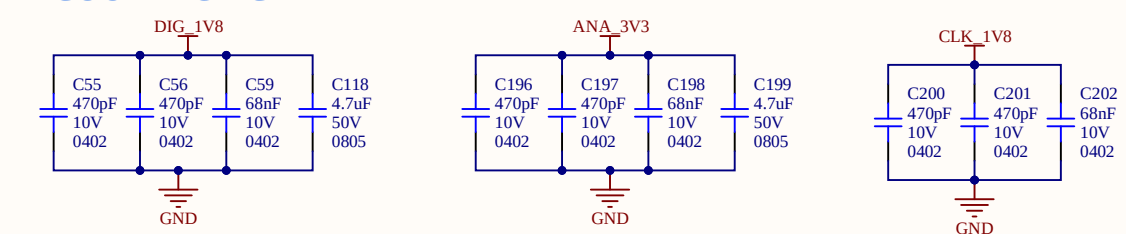
CNA (DAC)



DECOUPLAGE AMPLI



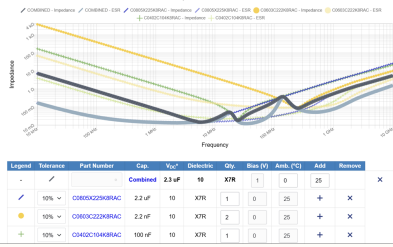
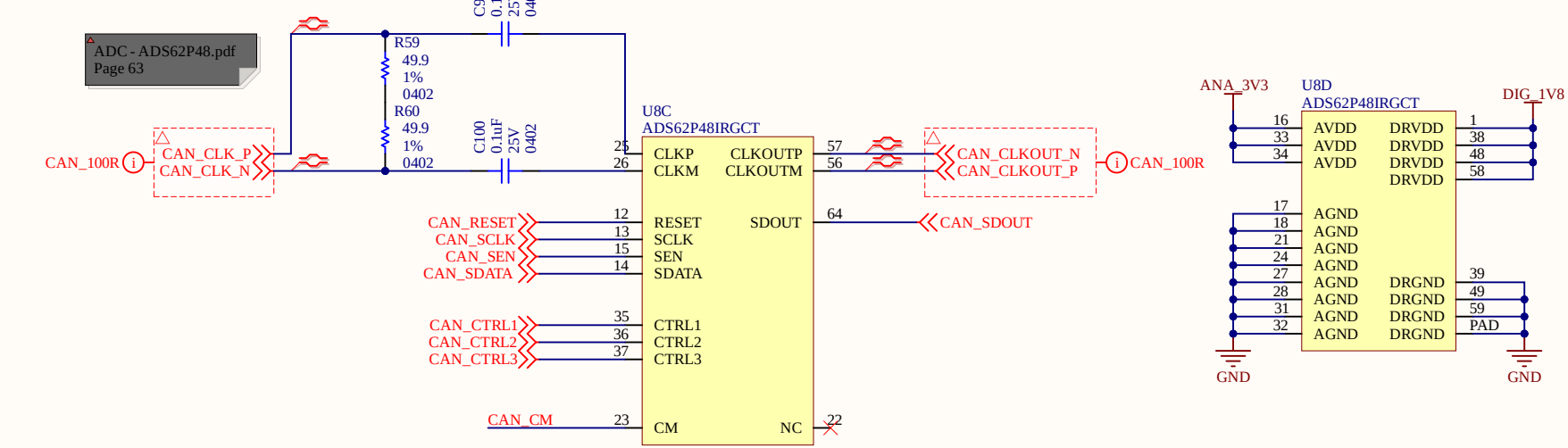
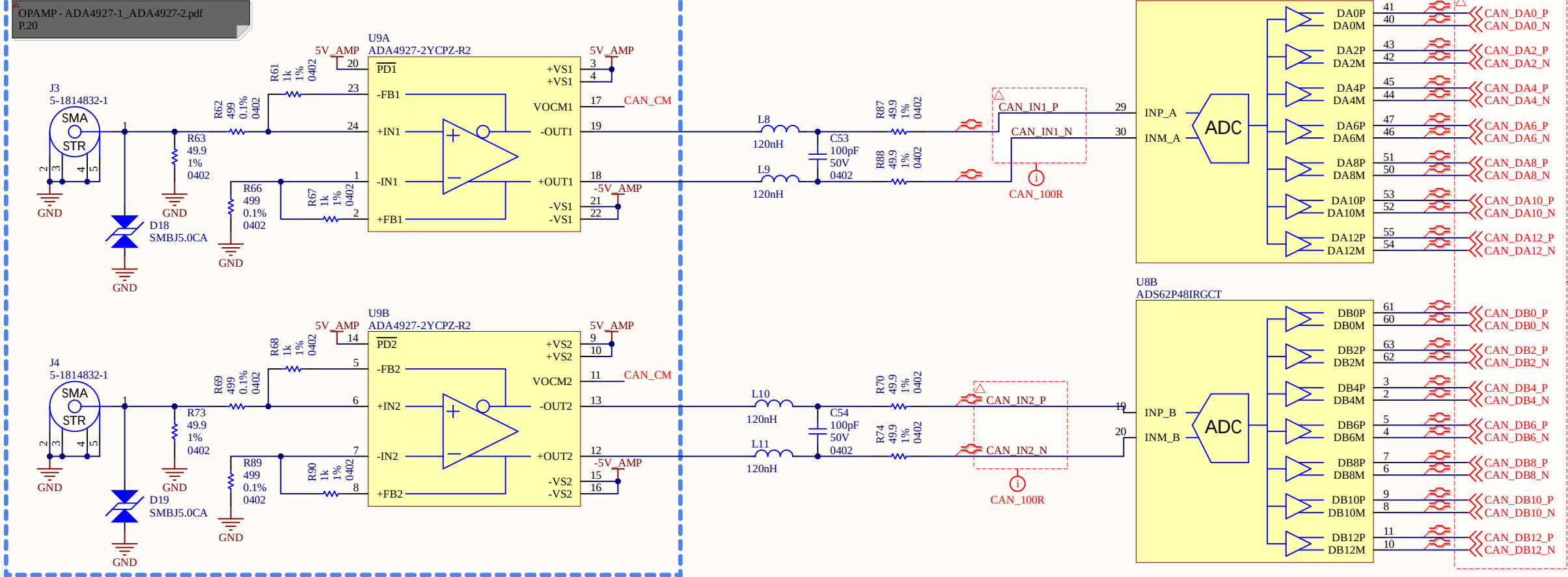
DECOUPLAGE CNA



-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On réussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C
-Receiver Differential impedance 80 to 120ohms

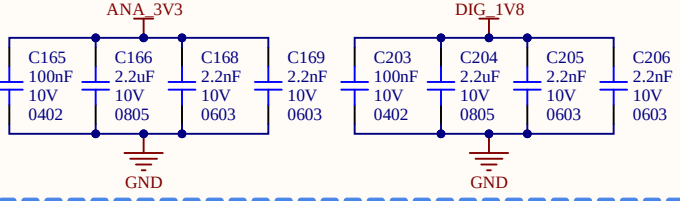
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Global Project		Pleiades		
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Date	13/01/2026		Sheet	6 of 17
Filename		S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc		
		Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CAN (ADC)
DECOUPLAGE ANALOG

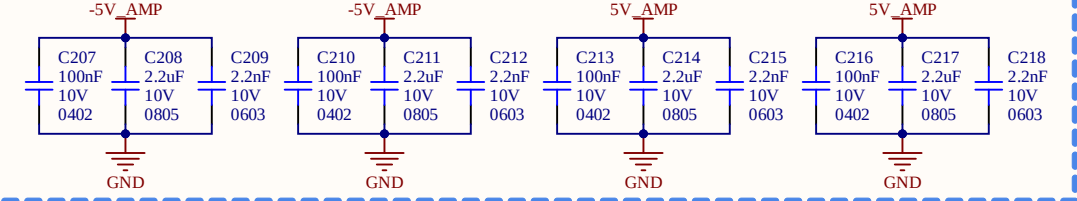


-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC

DECOUPLAGE CAN



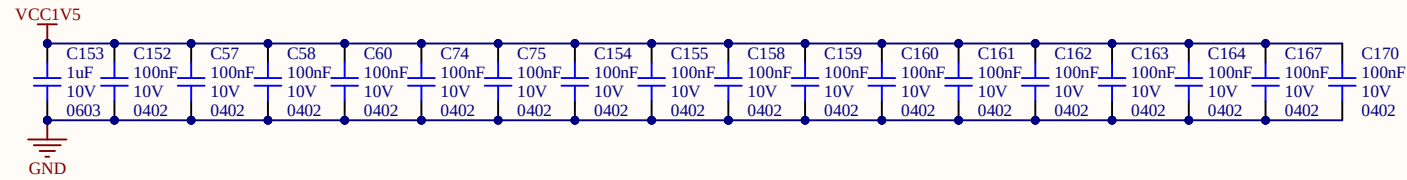
DECOUPLAGE AMPLI



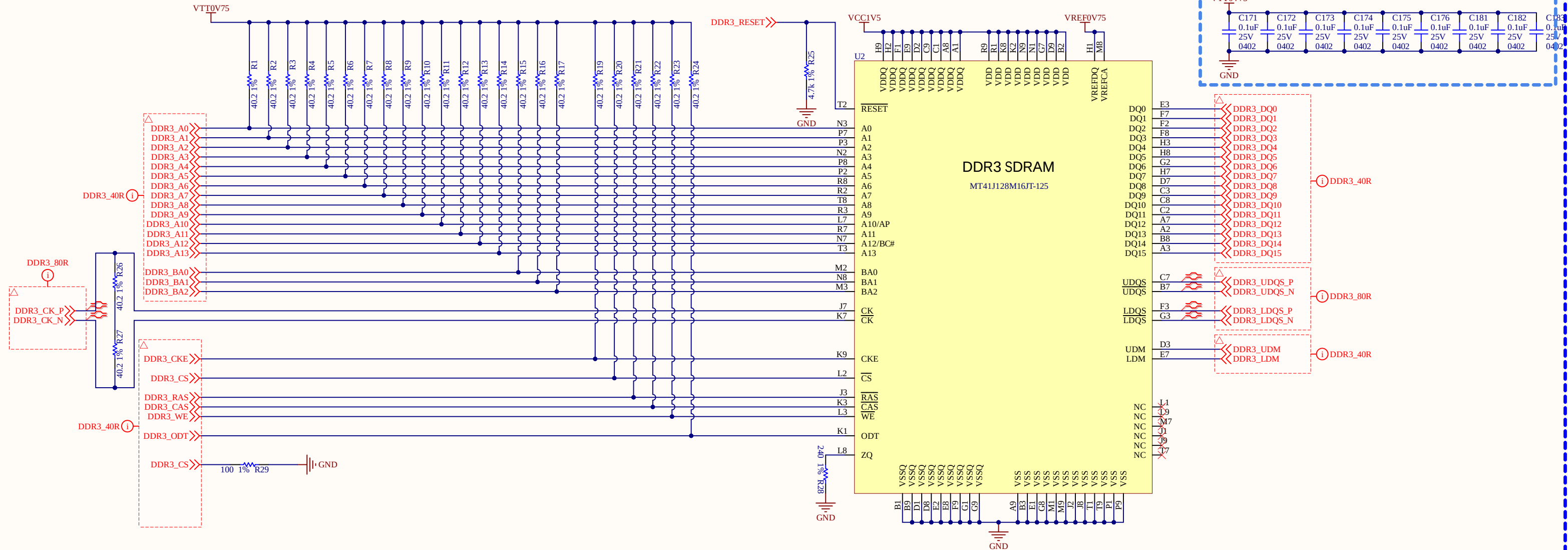
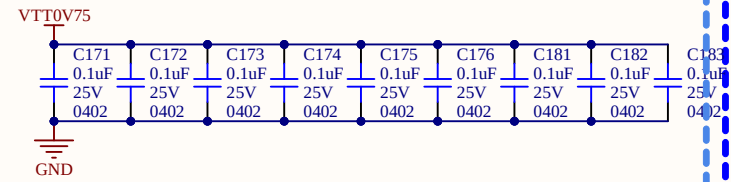
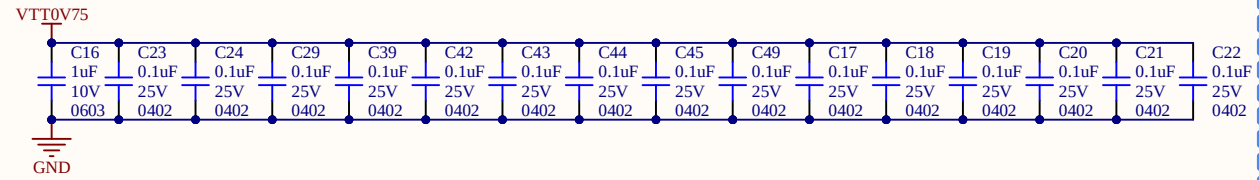
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Global Project		Pleiades	
Size	Group	Revision	
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Filename		Designers	
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FPGA

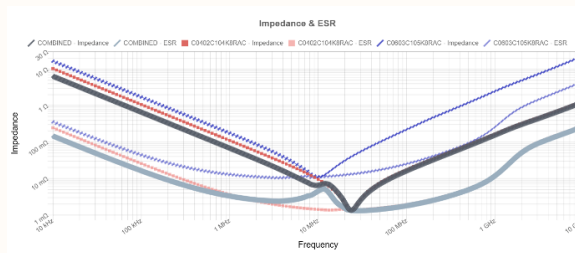
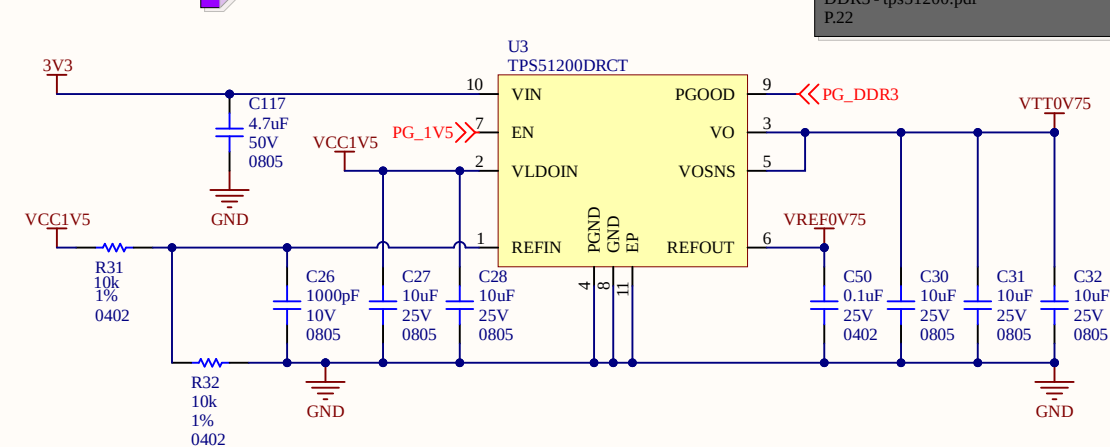
DECOUPLAGE



DECOUPLAGE



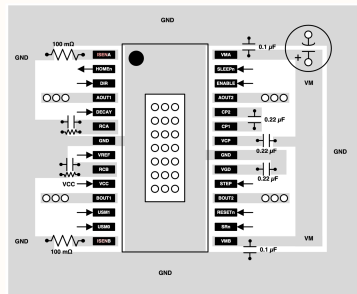
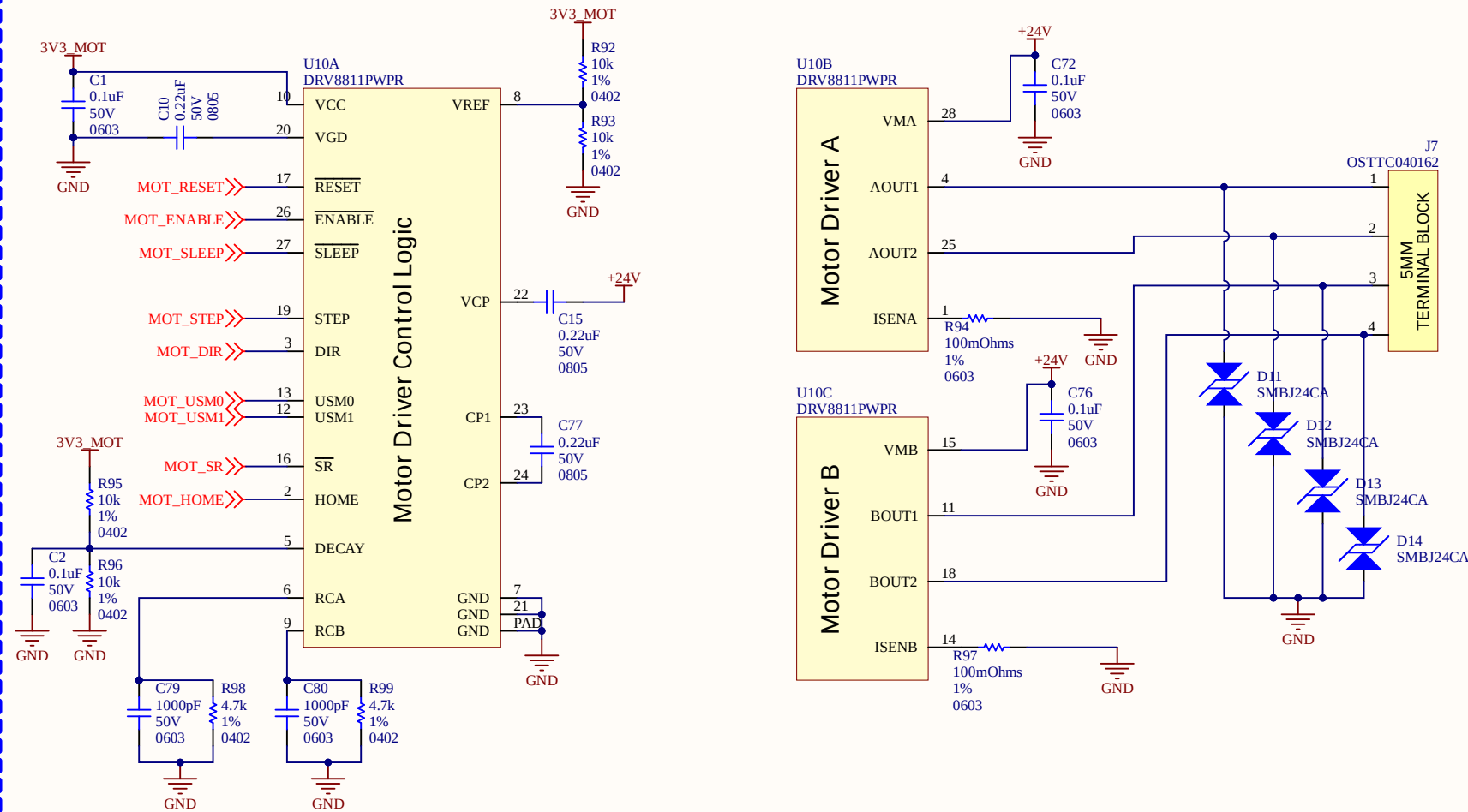
ALIM DDR3



-Ligne de transmission de 40R
-Resistance parallèle sur la ligne de transmission de 40.20ohms
-Clock en pair différentiel 80R
-Découplage il y a 18 pins donc on peut en mettre 18 condensateurs
-Dans un cours on s'est fait dire que c'est un condensateur par résistance de VTT

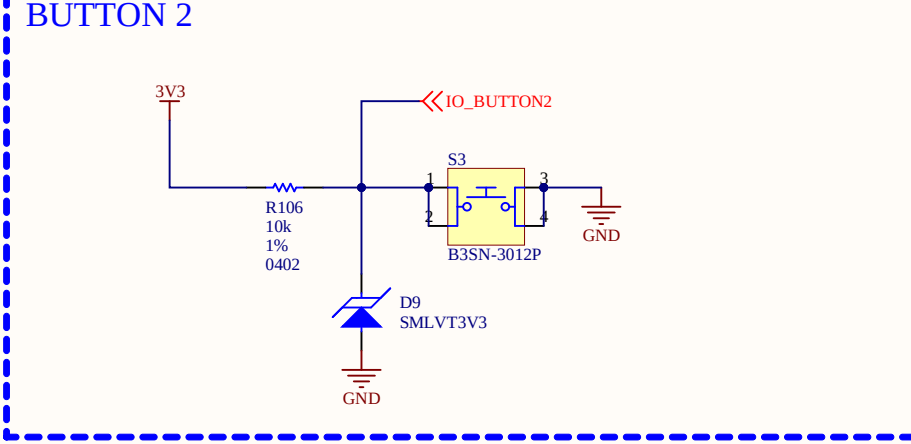
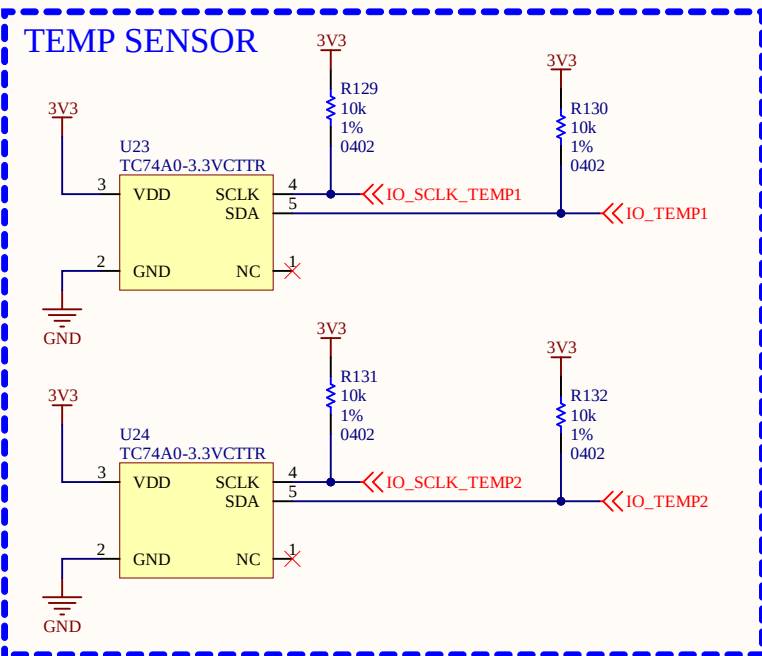
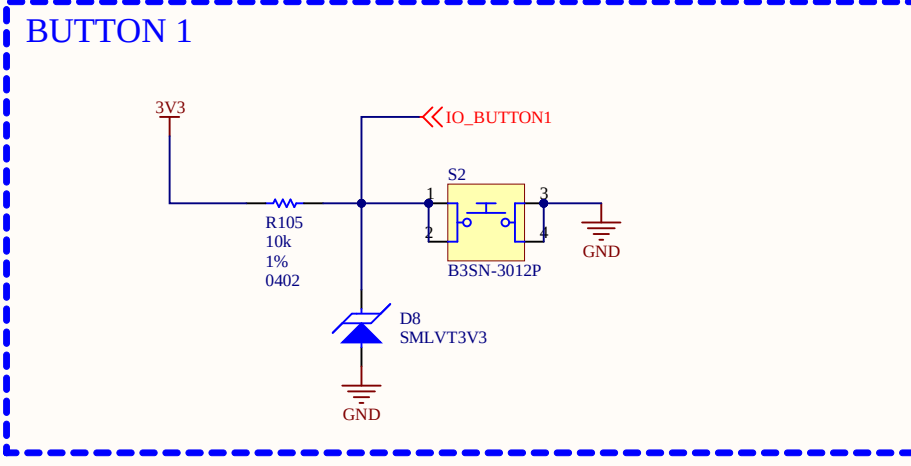
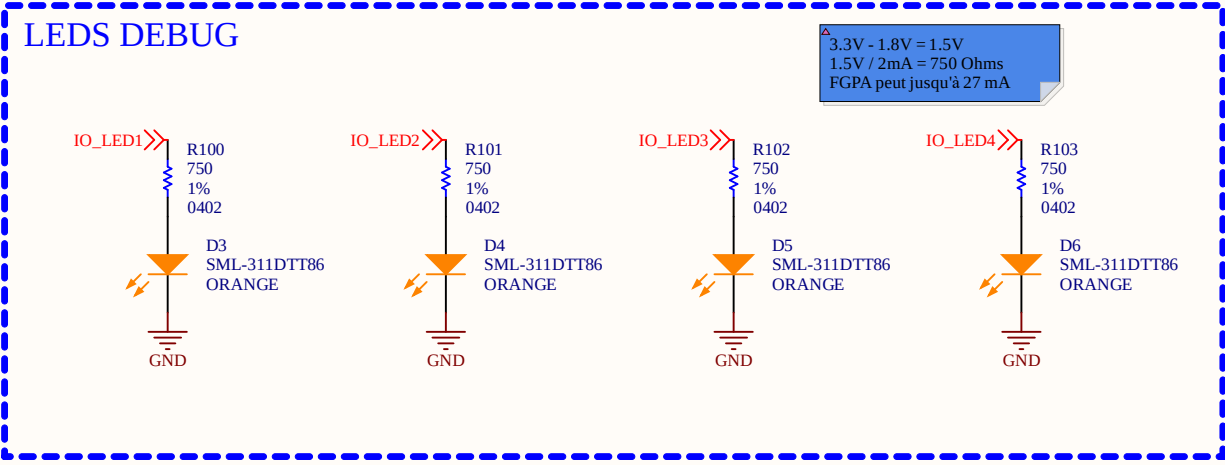
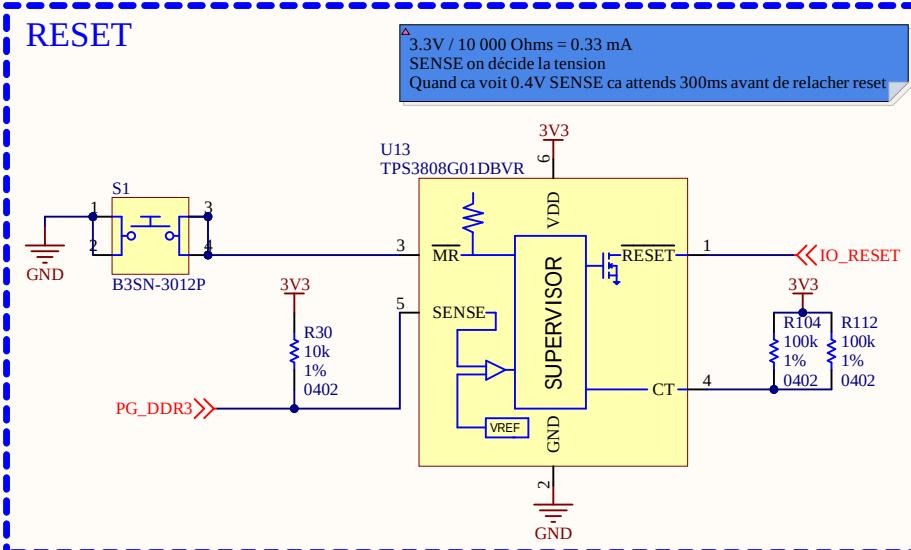
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Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

MOTOR



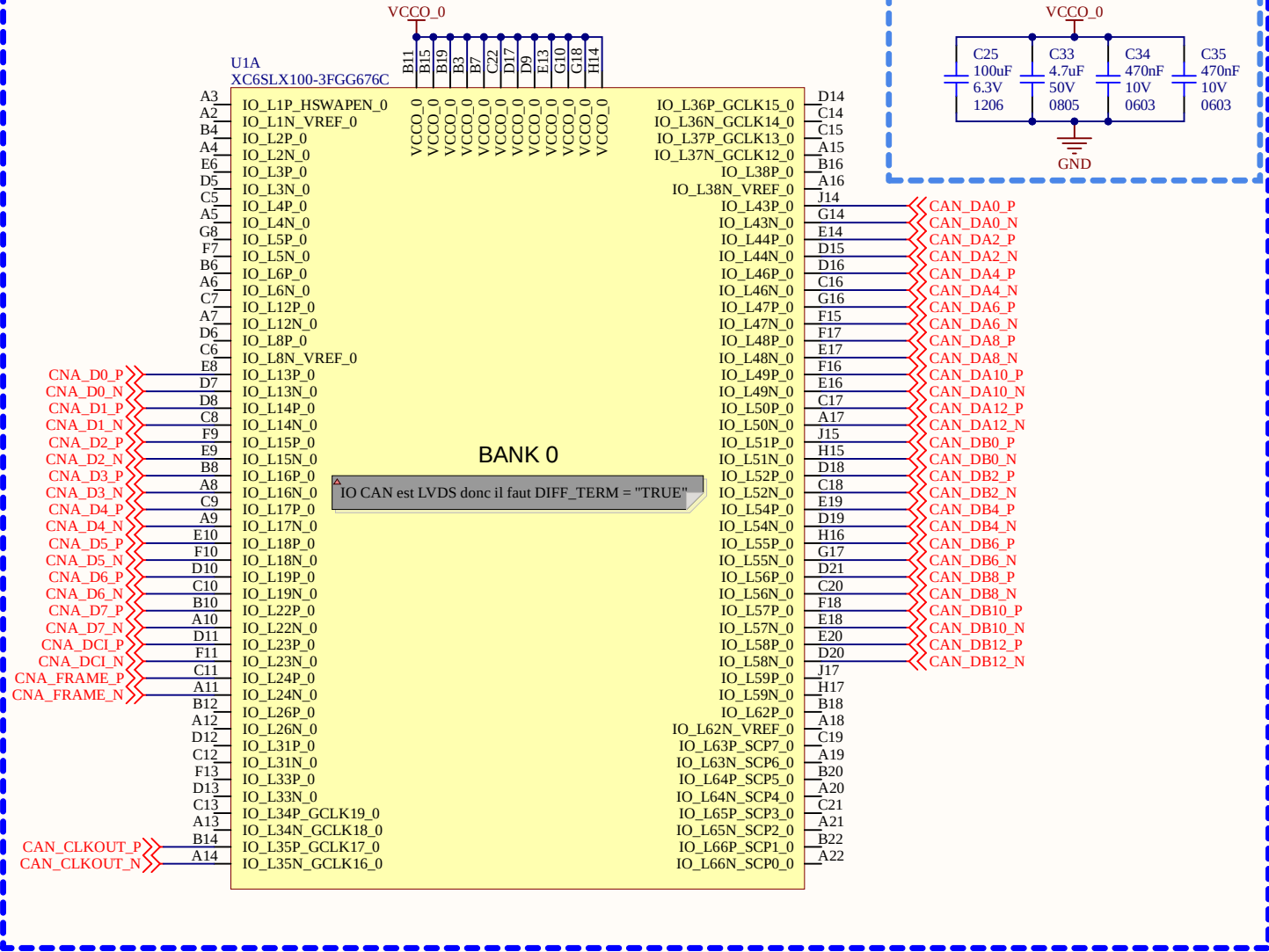
- Tous sur la BANK1
- ISENS = $1.65 / (8 \times 1,9A) = 0,108 \text{ Ohms}$

Sheet Name			MOTORS		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date		Sheet		of	
13/01/2026		9		17	
Filename			Designers		
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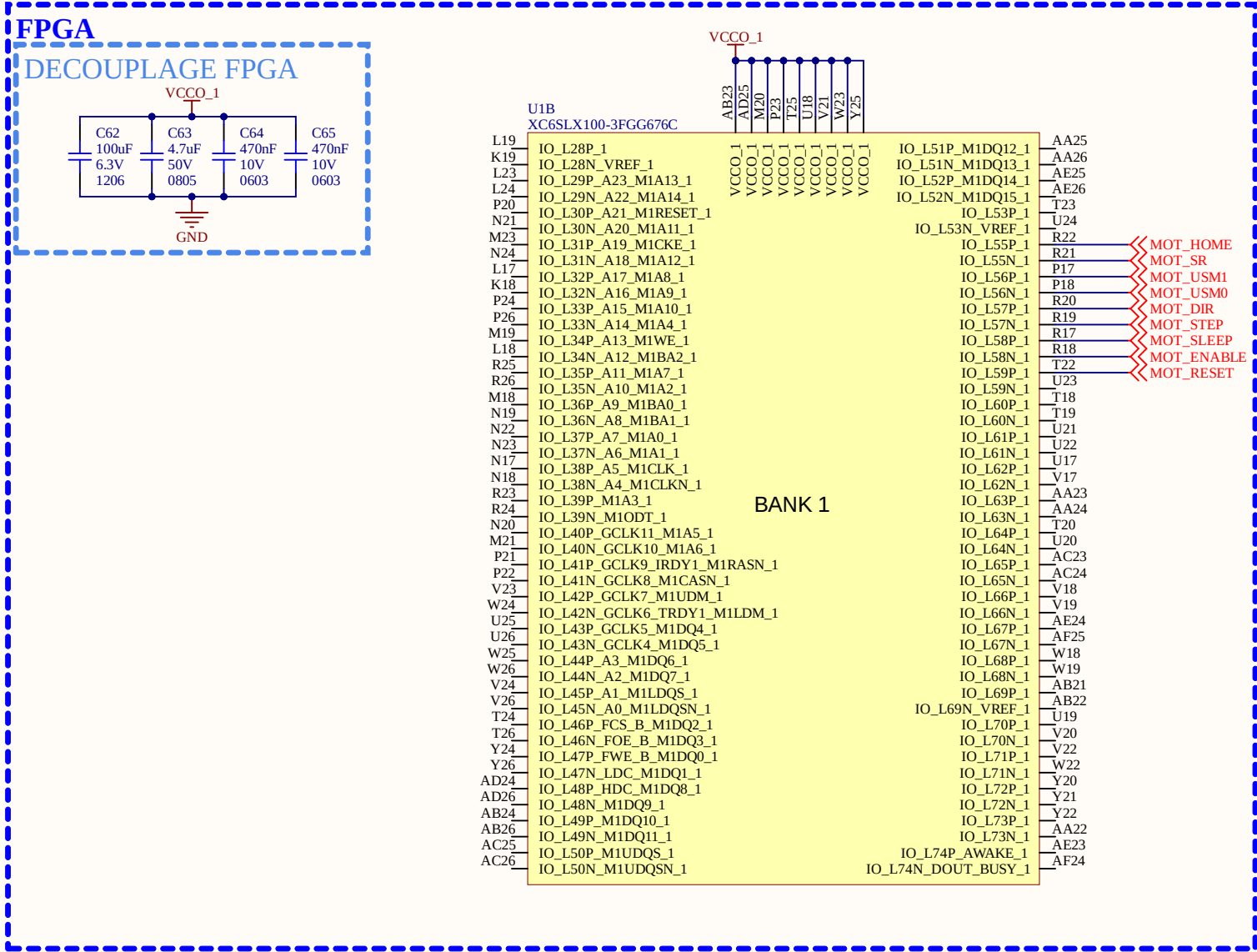
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Global Project				Pleiades			
Size		Group			Revision		
11x17		01			1.00		
Date			Sheet				
13/01/2026			10 of 17				
Filename					Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_IO.SchDoc					Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

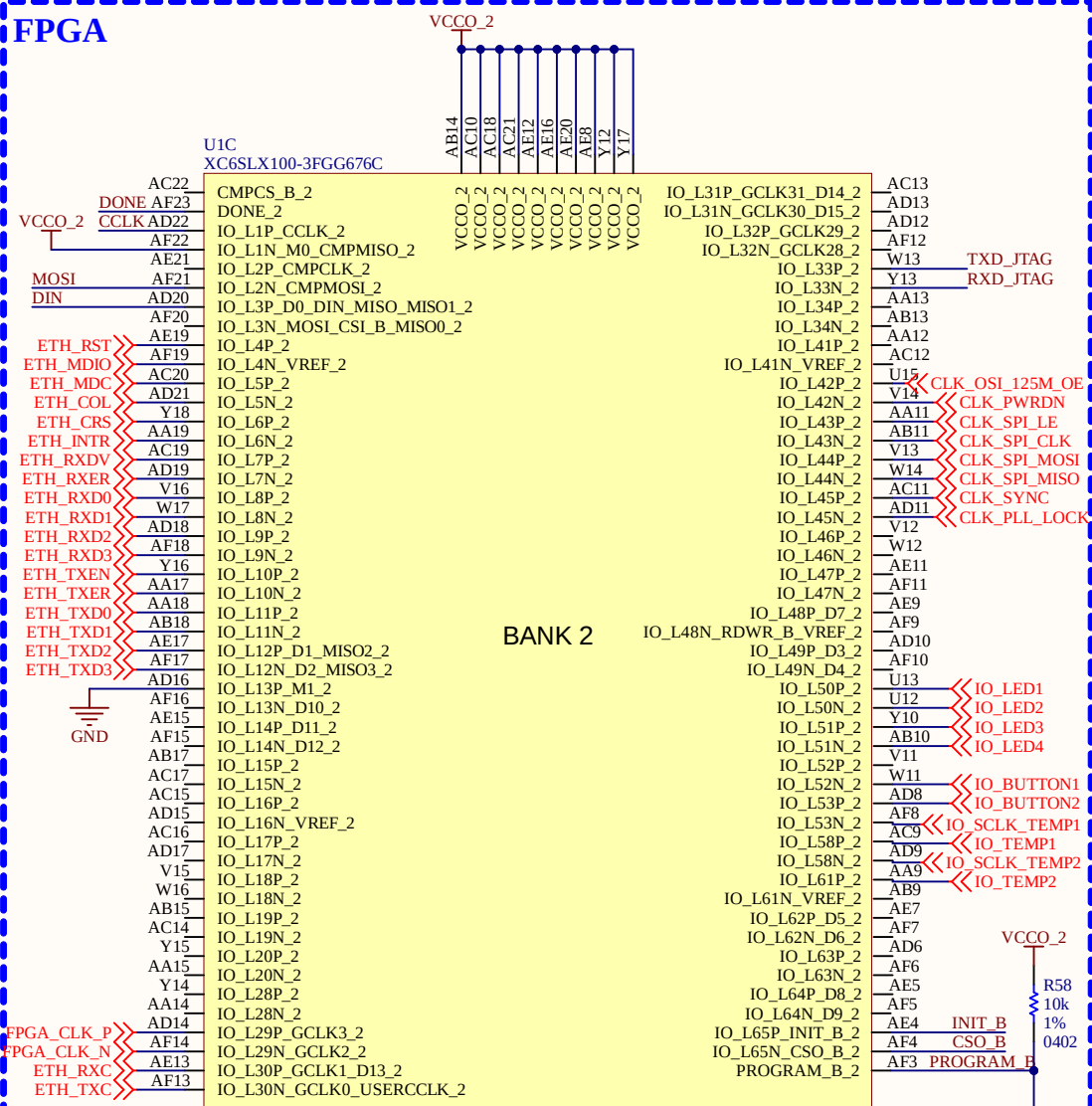


-LVDS seulement sur BANK0 et BANK2
-Toute les CAN et CNA sont mit sur cette BANK

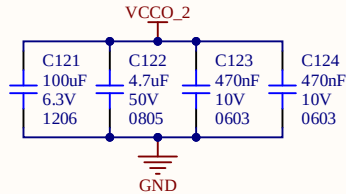
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Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	11	of 17
Filename				Designers	
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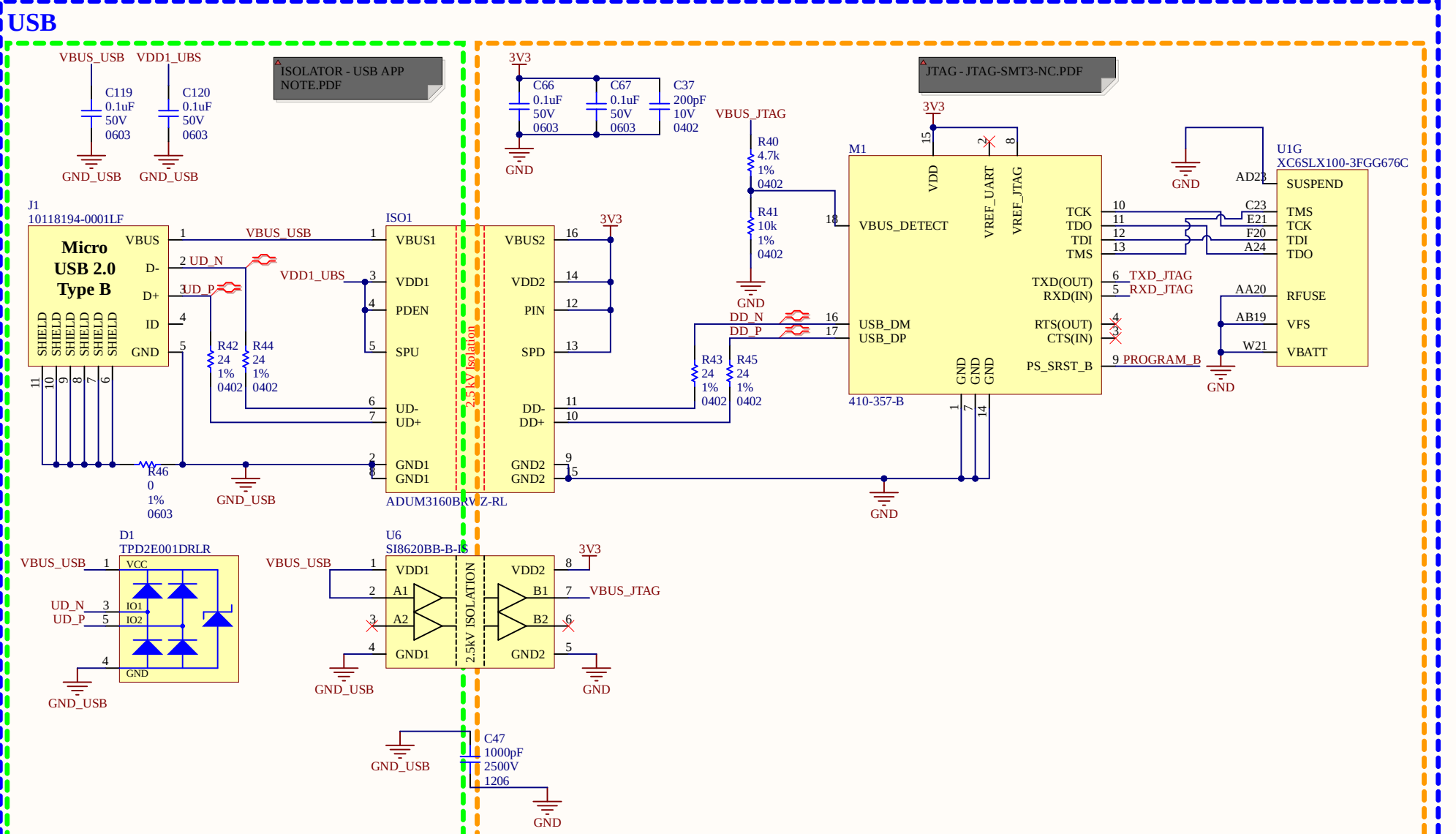
FPGA



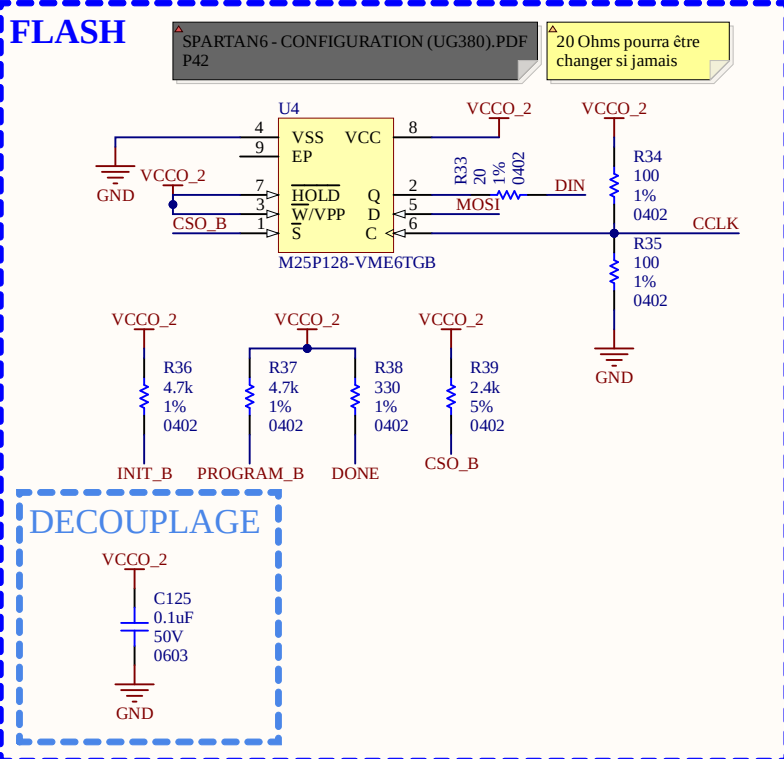
DECOUPLAGE FPGA



USB



FLASH

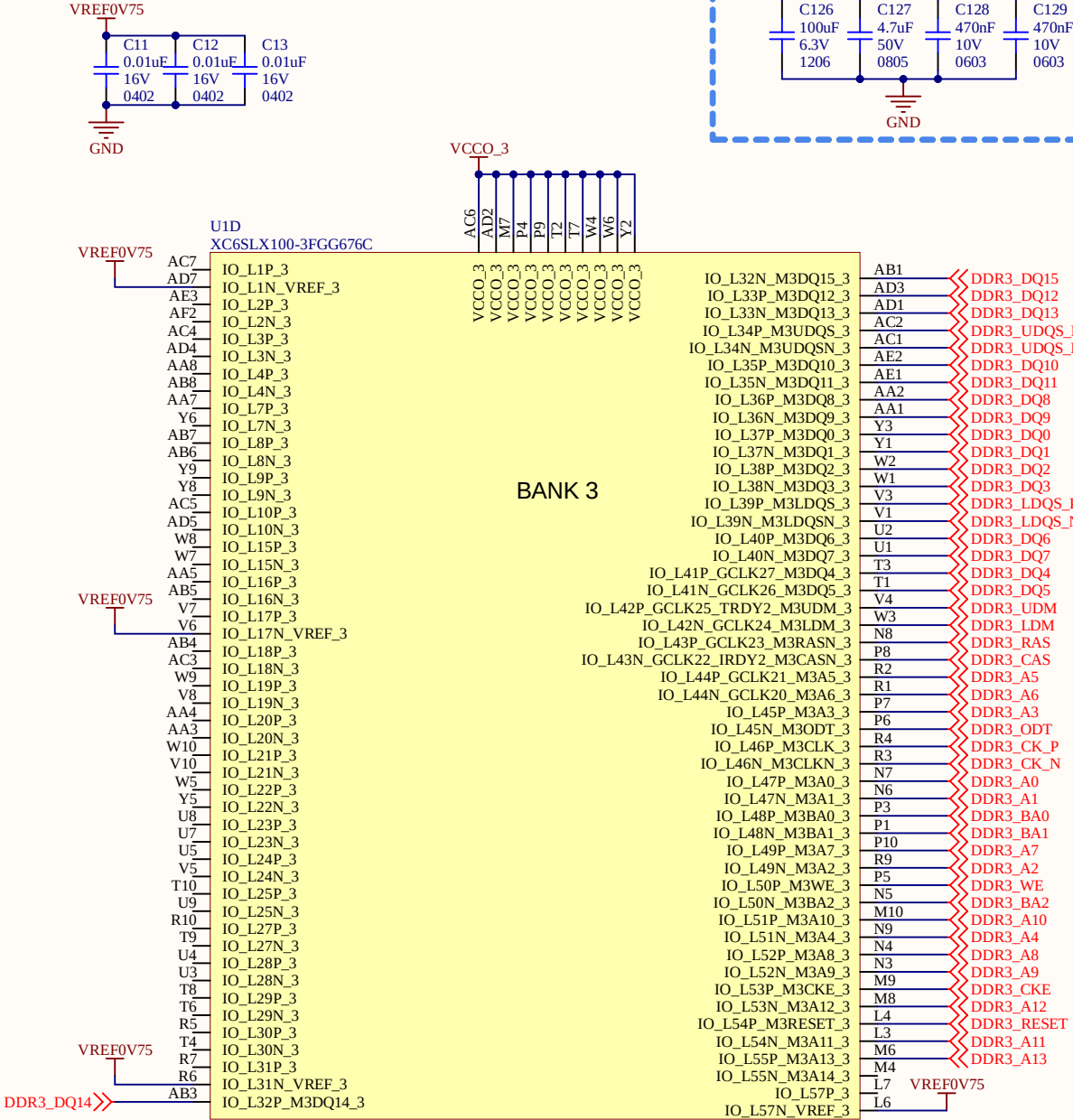


- Ajout de ESD pour la protection
 - Protection isolé pour le VBUS
 - Condensateur 2.5kV pour isolé les GND (même force que les isolateurs)
 - Paires différentiel sur le D+ et D-, avec des résistance de 240ohms (comme l'exemple)
 - Isolation du côté USB au lieu d'être du côté board (plus petit comme chip à comparer du ISO de 6 fils)
 - If AES encryption is unused, VBATT can be tied to either, VCCAUX or ground, or left unconnected.
 - VFS and RFUSE are tied to ground because we are not using Efuses
-
- M0 et M1 choisi pour avoir le mode SPI
 - Ajout d'une résistance de 20 ohms sur Q
 - Toute la logique de code est sur la BANK ainsi que toute la programmation

Sheet Name		FPGA - BANK 2	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	13 of 17
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

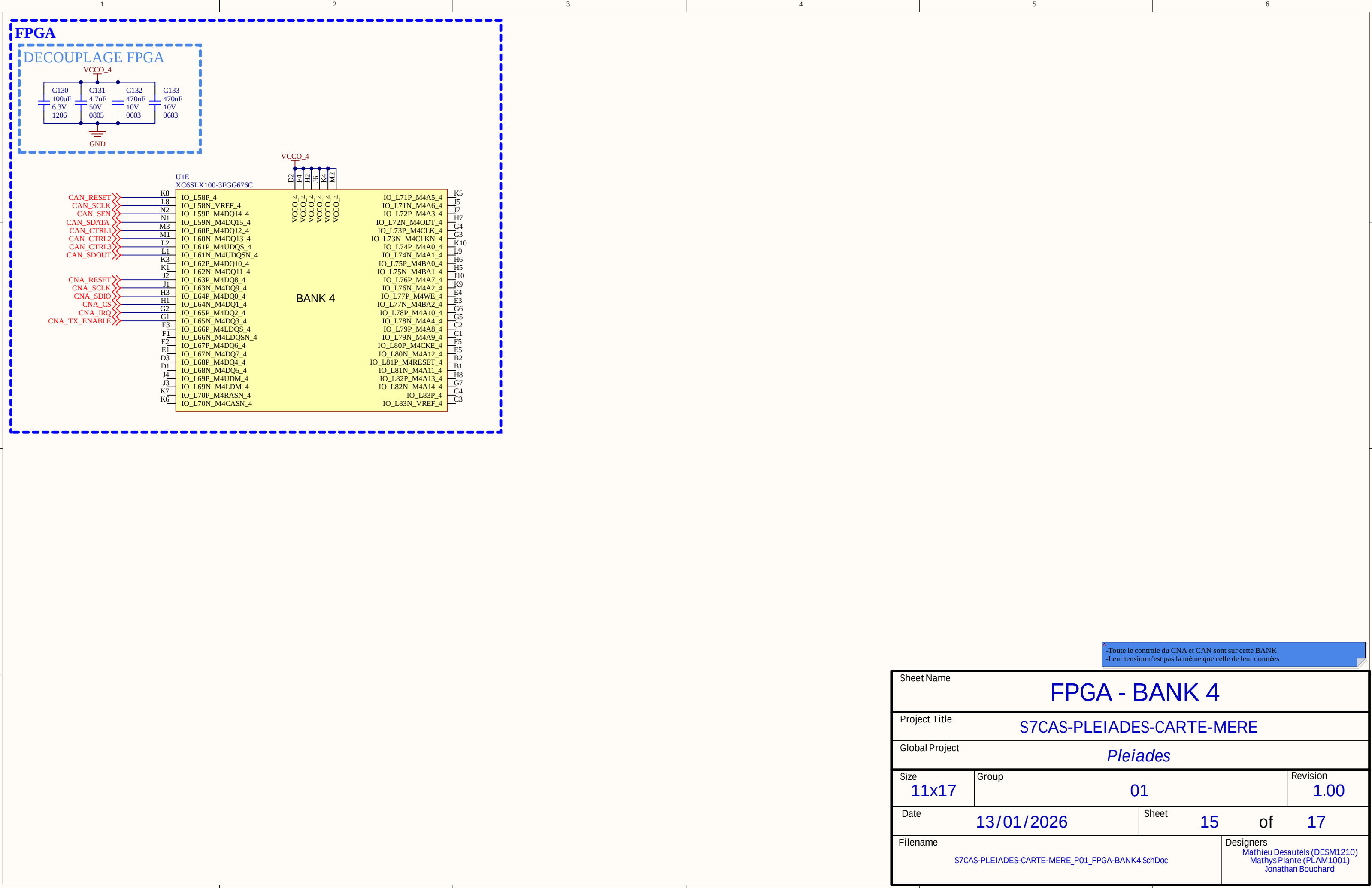
FPGA

DECOUPLAGE FPGA



-Toute la DDR3 est sur cette BANK

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
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Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



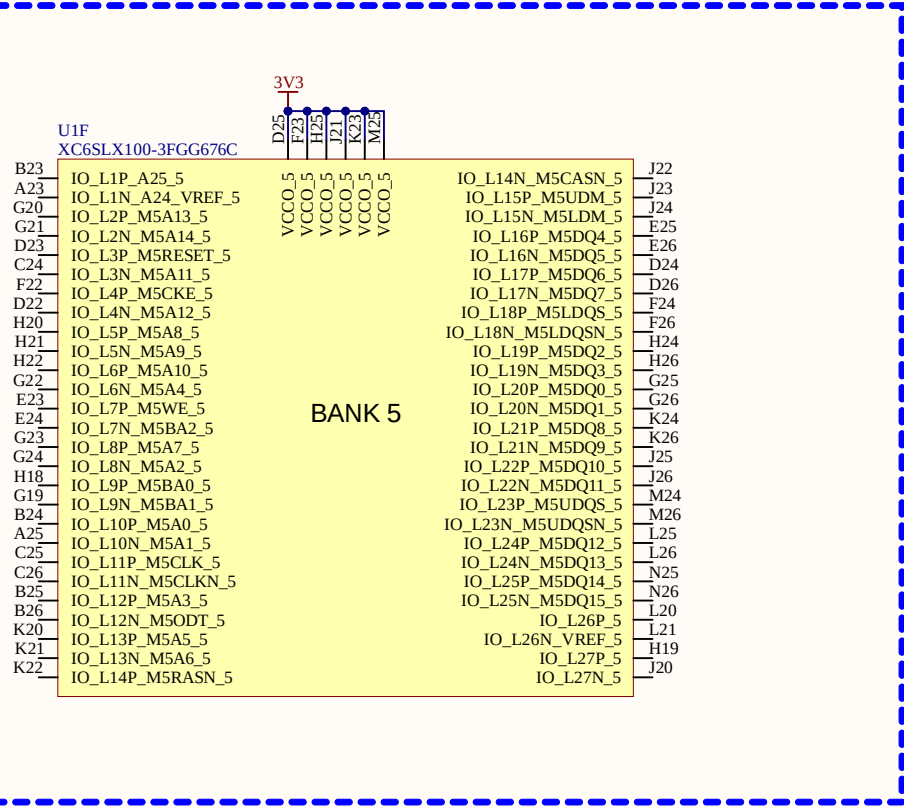
-Toute le controle du CNA et CAN sont sur cette BANK
-Leur tension n'est pas la même que celle de leur données

Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date		Sheet		of	
13/01/2026		15		17	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

3V3

Bank 5

Pin	Signal	Pin	Signal
B23	IO_L1P_A25_5	J22	IO_L14N_M5CASN_5
A23	IO_L1N_A24_VREF_5	J23	IO_L15P_M5UDM_5
G20	IO_L2P_M5A13_5	J24	IO_L15N_M5LDM_5
G21	IO_L2N_M5A14_5	E25	IO_L16P_M5DQ4_5
D23	IO_L3P_M5RESET_5	E26	IO_L16N_M5DQ5_5
C24	IO_L3N_M5A11_5	D24	IO_L17P_M5DQ6_5
F22	IO_L4P_M5CKE_5	D26	IO_L17N_M5DQ7_5
D22	IO_L4N_M5A12_5	F24	IO_L18P_M5LDQS_5
H20	IO_L5P_M5A8_5	F26	IO_L18N_M5LDQSN_5
H21	IO_L5N_M5A9_5	H24	IO_L19P_M5DQ2_5
H22	IO_L6P_M5A10_5	H26	IO_L19N_M5DQ3_5
G22	IO_L6N_M5A4_5	G25	IO_L20P_M5DQ0_5
E23	IO_L7P_M5WE_5	G26	IO_L20N_M5DQ1_5
E24	IO_L7N_M5BA2_5	K24	IO_L21P_M5DQ8_5
G23	IO_L8P_M5A7_5	K26	IO_L21N_M5DQ9_5
G24	IO_L8N_M5A2_5	J25	IO_L22P_M5DQ10_5
H18	IO_L9P_M5BA0_5	J26	IO_L22N_M5DQ11_5
G19	IO_L9N_M5BA1_5	M24	IO_L23P_M5UDQ5_5
B24	IO_L10P_M5A0_5	M26	IO_L23N_M5UDQSN_5
A25	IO_L10N_M5A1_5	L25	IO_L24P_M5DQ12_5
C25	IO_L11P_M5CLK_5	L26	IO_L24N_M5DQ13_5
C26	IO_L11N_M5CLKN_5	N25	IO_L25P_M5DQ14_5
B25	IO_L12P_M5A3_5	N26	IO_L25N_M5DQ15_5
B26	IO_L12N_M5ODT_5	L20	IO_L26P_5
K20	IO_L13P_M5A5_5	L21	IO_L26N_VREF_5
K21	IO_L13N_M5A6_5	H19	IO_L27P_5
K22	IO_L14P_M5RASN_5	J20	IO_L27N_5



Sheet Name			FPGA - BANK 5		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	16	of	17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK5.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

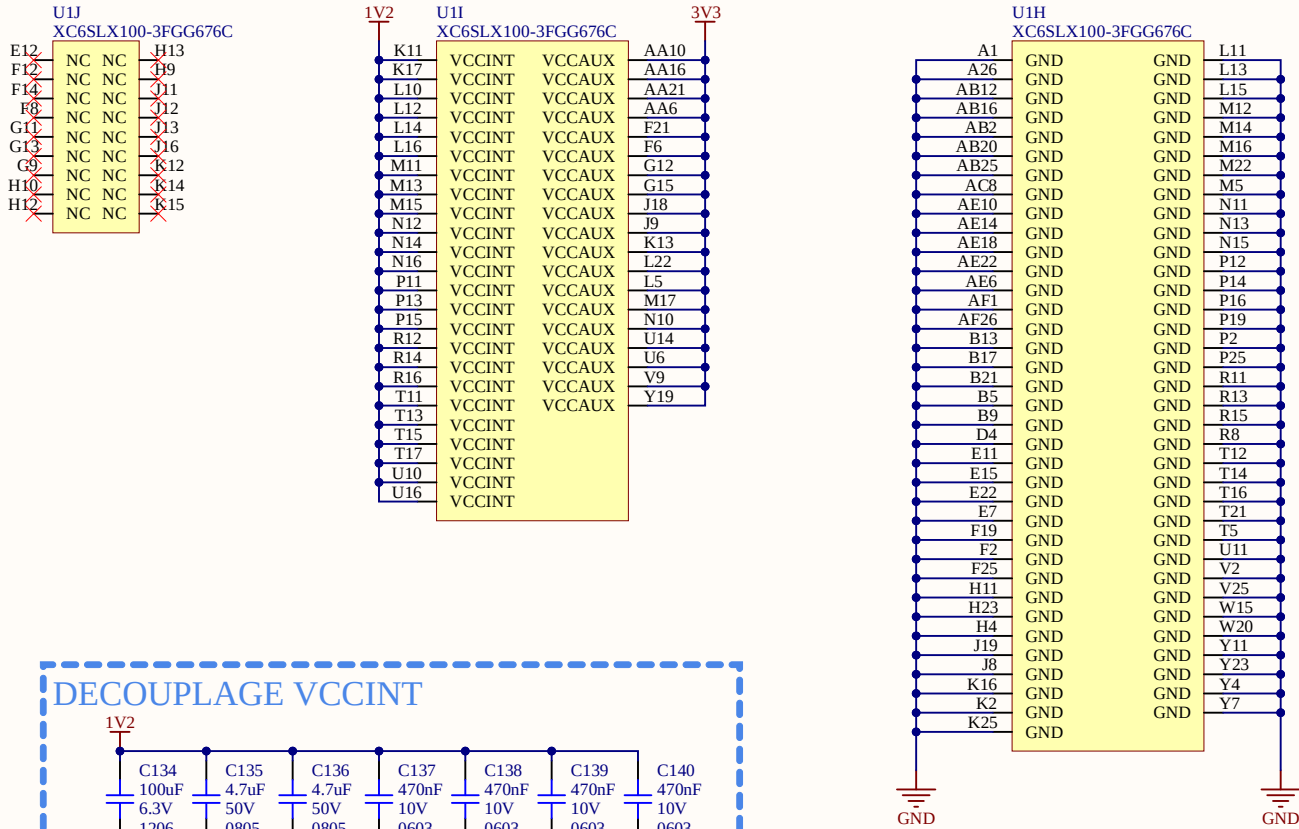


Table 2-1: Required PCB Capacitor Quantities per Device ⁽¹⁾⁽⁹⁾ (Continued)										
Package	Device (XC6S)	VCCINT in µF	VCCAUX in µF	VCC0 Bank 0 in µF	VCC0 Bank 1 in µF	VCC0 Bank 2 in µF	VCC0 Bank 3 in µF	VCC0 Bank 4 in µF	VCC0 Bank 5 in µF	Total (9)
		100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	100 4.7 0.47	
FG(G)484 LX75		1 2 3	1 2 4	1 1 2	1 1 3	1 1 4	1 1 3			33
FG(G)484 LX75T		1 2 3	1 2 4	1 1 2	1 1 3	1 1 3	1 1 3			32
FG(G)484 LX100		1 2 4	1 2 4	1 1 2	1 1 3	1 1 3	1 1 3			33
FG(G)484 LX100T		1 2 4	1 2 4	1 1 2	1 1 3	1 1 3	1 1 3			33
FG(G)484 LX150		2 3 6	1 2 4	1 1 2	1 1 3	1 1 3	1 1 3			37
FG(G)484 LX150T		2 3 6	1 2 4	1 1 2	1 1 3	1 1 3	1 1 3			37
FG(G)676 LX45		1 1 2	1 2 5	1 1 3	1 1 3	1 1 3	1 1 4			33
FG(G)676 LX75		1 2 3	2 3 6	1 1 3	1 1 3	1 1 3	1 1 3	1 1 2	1 1 2	45
FG(G)676 LX75T		1 2 3	1 2 5	1 1 3	1 1 2	1 1 3	1 1 2	1 1 2	1 1 2	40
FG(G)676 LX100		1 2 4	2 3 6	1 1 3	1 1 3	1 1 3	1 1 3	1 1 2	1 1 2	46
FG(G)676 LX100T		1 2 4	1 2 5	1 1 3	1 1 2	1 1 3	1 1 2	1 1 2	1 1 2	41
FG(G)676 LX150		2 3 6	2 3 6	1 1 3	1 1 3	1 1 3	1 1 3	1 1 2	1 1 2	50
FG(G)676 LX150T		2 3 6	1 2 5	1 1 3	1 1 2	1 1 3	1 1 2	1 1 2	1 1 2	45
FG(G)900 LX100T		1 2 4	2 3 6	1 1 3	1 1 3	1 1 3	1 1 4	1 1 2	1 1 2	47
FG(G)900 LX150		2 3 6	2 4 7	1 1 5	1 1 3	1 1 5	1 1 4	1 1 2	1 1 2	57
FG(G)900 LX150T		2 3 6	2 3 7	1 1 4	1 1 3	1 1 4	1 1 4	1 1 2	1 1 2	54

Sheet Name		FPGA - ALIM	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	17 of 17
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-ALIM.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	